WORKED SOLUTION

STEP III 1990, Question 8

Pure • Exact Differential Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q8

The key idea

Expand $\dfrac{\mathrm{d}}{\mathrm{d}x}(py' + qy)$ and compare it with $Py'' + Qy' + Ry$: this pins down p and q completely, and the condition $P'' - Q' + R = 0$ is exactly what makes the last coefficient fit. For the differential equation, the condition holds, so one integration takes us straight from a second-order equation to a first-order linear one, which an integrating factor then finishes.

Solution

The exactness criterion

For any functions p, q and y, the product rule gives

$$\frac{\mathrm{d}}{\mathrm{d}x}(py' + qy) = py'' + (p' + q)y' + q'y. \qquad (*)$$

Only if. Suppose p and q (which don't depend on y) make $Py'' + Qy' + Ry$ equal to $(*)$ for every y. Taking $y = 1$ gives $R = q'$. Taking $y = x$ gives $Q + Rx = (p' + q) + q'x$, so $Q = p' + q$. Taking $y = \frac{1}{2}x^2$ gives $P + Qx + \frac{1}{2}Rx^2 = p + (p' + q)x + \frac{1}{2}q'x^2$, so $P = p$. Therefore

$$P'' - Q' + R = p'' - (p'' + q') + q' = 0.$$

If. Suppose $P'' - Q' + R = 0$. Choose

$$p = P, \qquad q = Q - P'.$$

Then $p' + q = P' + Q - P' = Q$ and $q' = Q' - P'' = R$, using the condition. By $(*)$, $\dfrac{\mathrm{d}}{\mathrm{d}x}(py' + qy) = Py'' + Qy' + Ry$ for every y, as required.

The working also shows that p and q are forced to be P and $Q - P'$, which is how we'll find them below.

The differential equation

Here $P = x - x^4$, $Q = 1 - 7x^3$ and $R = -9x^2$, so

$$P'' - Q' + R = -12x^2 - (-21x^2) - 9x^2 = 0.$$

The criterion applies, with $p = P = x - x^4$ and $q = Q - P' = (1 - 7x^3) - (1 - 4x^3) = -3x^3$. The equation becomes

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[(x - x^4)y' - 3x^3 y\right] = (x^3 + 3x)\mathrm{e}^x.$$

Integrating by parts repeatedly, $\int x^3 \mathrm{e}^x \,\mathrm{d}x = (x^3 - 3x^2 + 6x - 6)\mathrm{e}^x$ and $\int 3x\mathrm{e}^x \,\mathrm{d}x = (3x - 3)\mathrm{e}^x$, so

$$(x - x^4)y' - 3x^3 y = (x^3 - 3x^2 + 9x - 9)\mathrm{e}^x + C.$$

At $x = 2$ we have $y = 2\mathrm{e}^2$ and $y' = 0$, so the left side is $-3 \cdot 8 \cdot 2\mathrm{e}^2 = -48\mathrm{e}^2$ and the right side is $5\mathrm{e}^2 + C$. Hence $C = -53\mathrm{e}^2$.

? **Ask yourself** The left side $(x - x^4)y' - 3x^3 y$ isn't the derivative of $(x - x^4)y$. How do we integrate a second time?

Factor out x: the left side is $x[(1 - x^3)y' - 3x^2 y] = x\,\dfrac{\mathrm{d}}{\mathrm{d}x}[(1 - x^3)y]$. (This is the usual integrating factor: $y' - \frac{3x^2}{1-x^3}y$ has factor $\exp\int \frac{-3x^2}{1-x^3}\,\mathrm{d}x = 1 - x^3$.) Dividing by x, which is allowed since the solution lives near $x = 2$,

$$\frac{\mathrm{d}}{\mathrm{d}x}[(1 - x^3)y] = (x^2 - 3x + 9)\mathrm{e}^x - \frac{9\mathrm{e}^x}{x} - \frac{53\mathrm{e}^2}{x}.$$

Integrating by parts again, $\int (x^2 - 3x + 9)\mathrm{e}^x \,\mathrm{d}x = (x^2 - 5x + 14)\mathrm{e}^x$. The term e^x/x has no elementary antiderivative, so we keep it as an integral from the starting point $x = 2$:

$$(1 - x^3)y = (x^2 - 5x + 14)\mathrm{e}^x - 53\mathrm{e}^2 \ln\frac{x}{2} - 9\int_2^x \frac{\mathrm{e}^t}{t}\,\mathrm{d}t + D.$$

At $x = 2$ the left side is $-7 \cdot 2\mathrm{e}^2 = -14\mathrm{e}^2$ and the right side is $8\mathrm{e}^2 + D$, so $D = -22\mathrm{e}^2$. Therefore

$$\boxed{y = \frac{1}{1 - x^3}\left[(x^2 - 5x + 14)\mathrm{e}^x - 22\mathrm{e}^2 - 53\mathrm{e}^2 \ln\frac{x}{2} - 9\int_2^x \frac{\mathrm{e}^t}{t}\,\mathrm{d}t\right]}$$

for $x > 1$. The coefficient $x - x^4$ of y'' vanishes at $x = 1$, so the solution through $x = 2$ is defined on $x > 1$ and needn't extend past $x = 1$.

Checks

- At $x = 2$ the bracket is $8\mathrm{e}^2 - 22\mathrm{e}^2 = -14\mathrm{e}^2$, and dividing by $1 - 8 = -7$ gives $y = 2\mathrm{e}^2$, as required.

- Stepping the original equation numerically from $x = 2$ (fourth-order Runge–Kutta) gives $y(3) \approx 7.90306$, which matches the formula with the integral done by Simpson's rule. The two also agree at $x = 1.2$, 1.5 and 4.
- Expanding $\frac{\mathrm{d}}{\mathrm{d}x}\left[(x - x^4)y' - 3x^3y\right]$ gives back $(x - x^4)y'' + (1 - 7x^3)y' - 9x^2y$.

Common slips

- Forgetting the constant C after the first integration. It isn't zero here, and it produces the logarithm.
- Trying to force $\int \mathrm{e}^x / x \, \mathrm{d}x$ into closed form. It can't be done, so leave it as a definite integral with lower limit 2, which also makes the initial condition easy to apply.

Beyond STEP

An expression $Py'' + Qy' + Ry$ that is a perfect derivative is called *exact*. The expression $P'' - Q' + R$ is the left side of the *adjoint* equation $(Pz)'' - (Qz)' + Rz = 0$ evaluated at $z = 1$. In general, any solution z of the adjoint equation is an integrating factor that makes $z(Py'' + Qy' + Ry)$ exact. Adjoints lead on to self-adjoint (Sturm–Liouville) problems and Green's functions. Boyce and DiPrima, *Elementary Differential Equations*, covers exact and adjoint equations in its exercises on second-order linear equations.

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